INCH-POUND
MIL-DTL-18313H
10 December 2019
SUPERSEDING
MIL-S-18313G
11 April 1978

DETAIL SPECIFICATION

SLINGS, CARGO, NET, NYLON WEBBING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flexible, lightweight cargo net slings made of nylon webbing. This specification does not cover cargo net slings used for handling ordnance, weapons, and explosives (that includes underway replenishment missions) which are specified in NAVSEA Drawing 53711-8410930 (see 6.1).

1.2 Classification. Cargo net slings are of the following types, classes, and sizes, as specified (see 6.2):

1.2.1 Types. The types of nylon webbing cargo net slings are as follows:

a. Type I – Heavy duty, load capacity 4,500 pounds
b. Type II – Medium duty, load capacity 3,500 pounds
c. Type III – Light duty, load capacity 2,500 pounds

1.2.2 Classes. The classes of nylon webbing cargo net stings are as follows:

a. Class A – Anti-static treated (see 3.2.2.2)
b. Class B – Standard impregnation (see 3.2.2.3)

1.2.3 Sizes. The sizes of nylon webbing cargo net slings are as follows:

a. Size 1 – 10 feet by 10 feet (types I and II) (see figure 3)
b. Size 2 – 11 feet 8 inches by 11 feet 8 inches (type III) (see figure 4)
c. Size 3 – 12 feet by 12 feet (types I and II) (see figure 5)
d. Size 4 – 14 feet by 14 feet (types I and II) (see figure 6)
e. Size 5 – 14 feet 2 inches by 14 feet 2 inches (type III) (see figure 7)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3940

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59826 - Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-23223 - Webbing, Nylon, Slotted

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org/.)

ASTM INTERNATIONAL

ASTM A153/A153M - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at www.astm.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

SAE INTERNATIONAL

SAE HS-1086 - Metals & Alloys in the Unified Numbering System

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Production units. Production units shall be fabricated in quantities as specified (see 6.2) and shall be in accordance with the examination and tests specified in 4.2.

3.2 Material.

3.2.1 Hoist links. Hoist links shall be made from steel bars, American Iron and Steel Institute (AISI) No. 1020, 1022, 1023, 1025, 1030, or equal, normalized, in accordance with SAE HS-1086. After successfully completing the proof link test (see 4.6.2), links shall be galvanized by the hot-dip method.

3.2.1.1 Steel hoist links. Ends of the bar which form the steel hoist link shall be forged or welded together. The joint shall be in the straight section of the link. The finished link shall be smooth to prevent abrasion of the friction pads.

3.2.1.1.1 Welded links. Welded links, regardless of the type of weld joint, shall require radiographic inspection in accordance with procedural requirements of T9074-AS-GIB-010/271. Welds shall be in accordance with the class 3 requirements of MIL-STD-2035.

3.2.1.1.2 Forged links. Forged continuous links manufactured without a joint shall be excluded from radiographic inspection.

3.2.1.2 Dimensions of links. Links shall conform to the dimensions shown on figure 2. Industrial standard tolerances apply.

3.2.1.3 Galvanization of links. Hoisting links shall be galvanized in accordance with ASTM A153/A153M, class B-1.

3.2.2 Webbing. Webbing shall be nylon and of the slotted type in accordance with MIL-W-23223 and table I. The webbing used in the slings shall be free of splices.

TABLE I. Webbing used in slings.

Sling			Spacing of Slots	Webbing, MIL-W-23223	
Type	**Class**	**Size**	**Inches (±⅛)**	**Type**	**Class**
I	A	1, 3, 4	8	I	A
I	B	3	8	I	B
II	A	1, 3, 4	8	IV	A
II	B	3	8	IV	B
III	A	2, 5	10	VII	A

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

3.2.2.1 Color. The color of the webbing (excluding marker threads, web insertions, and anti-static treated webbing) shall be olive drab. Color of the web shall be as shown on figure 1 for the corresponding net type. Web insertions on class A nets shall not be treated with rubber latex.

3.2.2.2 Class A webbing. Class A webbing shall be anti-static treated with conductive rubber latex in accordance with MIL-W-23223 and shall result in charcoal color so that it is readily identified. This deposit shall increase the weight of the webbing by 7 to 14 percent. Web inserts for class A nets shall not be treated.

3.2.2.3 Class B webbing. Class B webbing shall be fire retardant webbing impregnated with neoprene latex (see 6.2). The treatment shall consist of impregnating the webbing with polychloroprene compound containing suitable acid acceptors and flame-proofing compounds in accordance with MIL-W-23223.

3.2.3 Thread. The thread shall be type I or type II, number size 4 or 6, or letter size FF in accordance with A-A-59826.

3.2.3.1 Color of thread. Thread may be either white or olive drab.

3.2.4 Manufacturer's identification. A label, in accordance with MIL-DTL-32075, shall be attached by sewing it to the frame webbing in each net corner. The label shall show the new manufacturer's name, month and year of manufacture, National Stock Number, capacity, and size of the net. Lettering on the label shall be at least ¼ inch high.

3.3 Construction. Type I and II slings shall be constructed as shown on figures 3, 5, and 6, as applicable. Type III slings shall be constructed as shown on figures 4 and 7, as applicable.

3.3.1 Meshes. Meshes shall be formed by passing the webbing alternately through webbing slot-loping an enclosing square pattern. The size of the meshes shall be established by the spacing of the slots; meshes shall be oriented at right angle to the frame. Insertions into a webbing shall be made by other webbings at spacings of 16 or 20 inches to form a square mesh pattern.

3.3.2 Frame. The frame of types I and II slings shall consist of four lengths of slotted webbings called frame webbings. Joining of the frame webbings shall be accomplished in two steps as shown on figures 3, 5, and 6. The frame of type III slings shall be of one piece as shown on figures 4 and 7.

3.3.3 Stitching. Stitches shall be in accordance with stitch type 301 of ASTM D6193 and shall be made continuous. When machine stitching is not a continuous thread, it shall be back-stitched a minimum of 1 inch at the ends to prevent raveling, except when ends are held down by other stitchings. Thread tension shall be properly maintained and there shall be no loose stitches.

3.3.3.1 Stitching pattern.

3.3.3.1.1 Number size 4. When number size 4 nylon thread is utilized, mesh intersections shall be as specified in detail B of figures 3 through 7. For type I and II slings, mesh intersections shall be stitched in an "X" pattern with two closed ends which shall be parallel to exposed edges of webbing or as an option a fully enclosed "X" pattern. There shall be 6 to 7 stitches to the inch.

3.3.3.1.2 Number size 6. When number size 6 nylon thread is utilized, mesh intersections shall be as specified in detail B of figures 3 through 7. For type I and II slings, mesh intersections shall be stitched in an "X" pattern with two closed ends which shall be parallel to exposed edges of the webbing or as an option a fully enclosed "X" pattern. There shall be 5 to 6 stitches to the inch.

3.3.3.1.3 Letter size FF. When letter size FF nylon thread is utilized, mesh intersections shall be as specified in detail B of figures 3 through 7. For type I and II slings, mesh intersections shall be stitched in a fully enclosed "X" pattern with two additional rows of stitching parallel to exposed edges of the webbing or as an option an additional box around the fully enclosed "X" pattern. There shall be 5 to 8 stitches to the inch.

3.3.3.1.4 Webbing. Frame webbing and superimposed mesh webbings shall be stitched along both edges of the webbings. Stitching at steel link corners shall go beyond the last web and as close to the steel link as practical without use of any special attachments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

3.3.4 Friction pads. Friction pads shall be used at each corner of type I and II slings to protect the frame webbing from abrasion by the hoist links. The pads shall be of the same type webbing used in the sling and shall be attached as shown on figures 3, 5, and 6. For type III slings, see note 4 on figures 4 and 7.

3.3.5 Size of slings. Each size of sling (see 1.2) shall be measured from center of the frame webbing. Dimensions may vary slightly due to minor tolerances of webbing slot spacings.

3.4 Workmanship. The workmanship of the completed slings shall be consistent with the rigging practice. The sling shall be free from any defects which may affect the serviceability. All cut ends of webbing shall be heat-sealed.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one test lot when a first article sample is required (see 3.1). This inspection shall include the examination outlined in 4.5 and the tests of 4.6.

4.3 Conformance inspection.

4.3.1 Sampling for conformance inspection.

4.3.1.1 Lot. Slings of the same type, class, and size presented for delivery at one time shall be considered a lot.

4.3.1.2 Sampling for examination of slings. A random sample of slings shall be selected from each lot in accordance with ASQ/ANSI Z1.4 at inspection level II.

4.3.1.3 Sampling of links for proof test. Sample links shall be selected in accordance with table II from the total quantity of links required for the slings in each lot, prior to galvanization and assembly with webbing for the test of 4.6.2. If any sample link fails in this test, this shall be cause for rejection of all links.

TABLE II. Sling links sample size for proof test.

Number of links	Sample size
4 to 16	2
20 to 40	3
44 to 68	5
72 to 112	7
116 to 300	10
304 to 500	15
504 to 800	25
804 to 1,300	35
1,304 to 3,200	50
3,204 to 8,000	75

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

4.4 Requirements and verification methods. Requirements of section 3 shall be inspected and verified according to the tests and examinations outlined in table III.

TABLE III. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Link material	X		3.2.1	4.5
Link construction	X		3.2.1.1	4.6.2
Link dimension	X	X	3.2.1.2	4.5
Link galvanization	X		3.2.1.3	4.5
Webbing material	X		3.2.2	4.5
Webbing color	X		3.2.2.1	4.5
Class A anti-static treatment	X		3.2.2.2	4.5
Class B fire retardant treatment	X		3.2.2.3	4.5
Thread material/size	X		3.2.3	4.5
Thread color	X		3.2.3.1	4.5
Manufacturer's identification	X	X	3.2.4	4.5
Sling construction	X	X	3.3	4.5
Mesh size and spacing	X	X	3.3.1	4.6.1
Frame attachment	X	X	3.3.2	4.5
Stitch type	X	X	3.3.3	4.5
Stitch pattern	X		3.3.3.1	4.5
Webbing stitching	X	X	3.3.3.1.4	4.5
Friction pad type	X		3.3.4	4.5
Friction pad attachment	X	X	3.3.4	4.5
Sling size	X	X	3.3.5	4.5
Sling defects	X	X	3.4	4.5

4.5 Examination of slings. Each of the sample slings selected in accordance with 4.3.1.2 shall be examined for defects classified as shown in table IV to determine compliance with the requirements of this specification which do not require tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Slings incomplete; parts missing.
102	Component parts old; evidence of reclaimed, used, or salvaged components.
103	Evidence of use of unauthorized materials or repairs.
104	Slings not size specified.
105	Hoist links ungalvanized.
106	Link surface not smooth; evidence of blisters, seams, laps, cracks, or pit holes.
107	Link configuration and dimensions nonconforming to figure 2.
108	Webbing not of specified type or has splices.
109	Integral slots of webbing not woven with specified length or spacing.
110	Webbing not impregnated with required rubber latex compound.
111	Nonconforming flame-proof treatment of webbing.
112	Evidence of dye, detergent, chemical, or finishing agent use, which may cause deterioration of webbing in normal storage.
113	Thread not made of specified nylon cord.
114	Nonconforming mesh configuration or dimensions.
115	Nonconforming frame configuration or composition.
116	Stitches not continuous, thread tension not properly maintained, or evidence of loose stitches.
117	Stitching pattern nonconforming: frame webbing and superimposed mesh webbings not stitched as specified.
118	Friction pads not same type of webbing, not located or attached as specified.
Minor	
201	Size identification web insertions not of assigned color; not inserted into body as specified.
202	Color of webbing nonconforming.

4.6 Tests.

4.6.1 Spacing of slots. Measurement of slot spacings shall be made on eight webbing samples. The exact centers of two consecutive slots on each sample shall be marked on the selvage, and the distance between marks measured with a metal tape graduated in 1/16 inch (see table I).

4.6.2 Link proof test. Each sample link, selected in accordance with 4.3.1.3, shall be in accordance with and comply to all dimensions of figure 2. Match marks shown on figure 2 shall be made and dimensions between them measured and recorded. The link shall then be subjected to a proof test of 2,225 pounds after load release. At the conclusion of the test, the link shall be examined for deformation. Any dimensional change from the recorded dimension shall be cause for rejection of the lot. The tested link shall be magnetic particle or liquid penetrant inspected in accordance with T9074-AS-GIB-010/271 and shall be in accordance with the class 3 acceptance standard of MIL-STD-2035.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Slings covered by this specification are intended primarily for loading and discharging general cargo, including use in underway replenishment missions. The slings may also be used as flexible staging, save-all nets, lash down slings, safety, man overboard, and abandon ship nets. Cargo net slings used for handling ordnance, weapons, and explosives (that includes underway replenishment missions) require additional testing and quality assurance provisions. These nets are now covered by NAVSEA Drawing 53711-8410930. For superseding information and additional requirements for these slings contact: Naval Surface Warfare Center, Indian Head Explosive Ordnance Disposal Technology Division (NSWCIHEODTD) Detachment Picatinny, Naval Packaging, Handling, Storage and Transportation (PHST) Center, Picatinny Arsenal, NJ.

6.2 Acquisition requirements. Acquisition documents should specify the following:

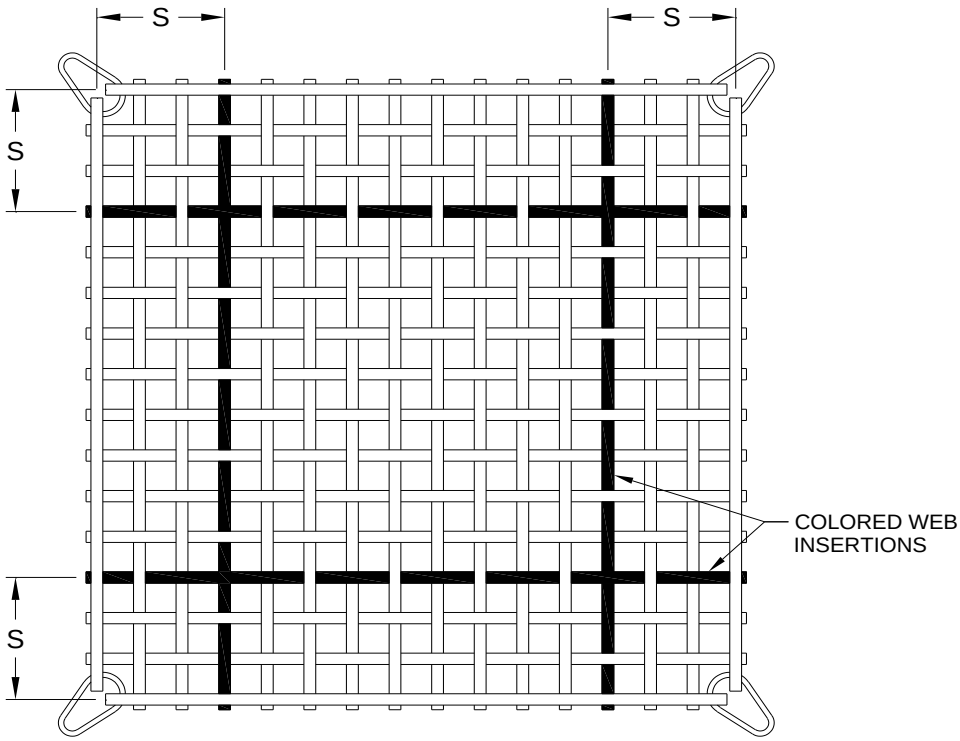
a. Title, number, and date of this specification.
b. Type, class, and size required (see 1.2).
c. When first article inspection is required (see 3.1).
d. Quantity of units (see 3.1.1).
e. Requirement for flameproof treatment of webbing (see 3.2.2.3).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Abandon ship
Anti-static
Flame proof
Frame
Man overboard
Mesh
Stitching

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.



Type of sling	Color of web insertion	S (Inches)				
		Sling size				
		1	2	3	4	5
I	Lt. Blue	32	-	32	40	-
II	Yellow	32	-	32	40	-
III	White	-	30	-	-	40

FIGURE 1. Color of nylon web insertions in cargo net slings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

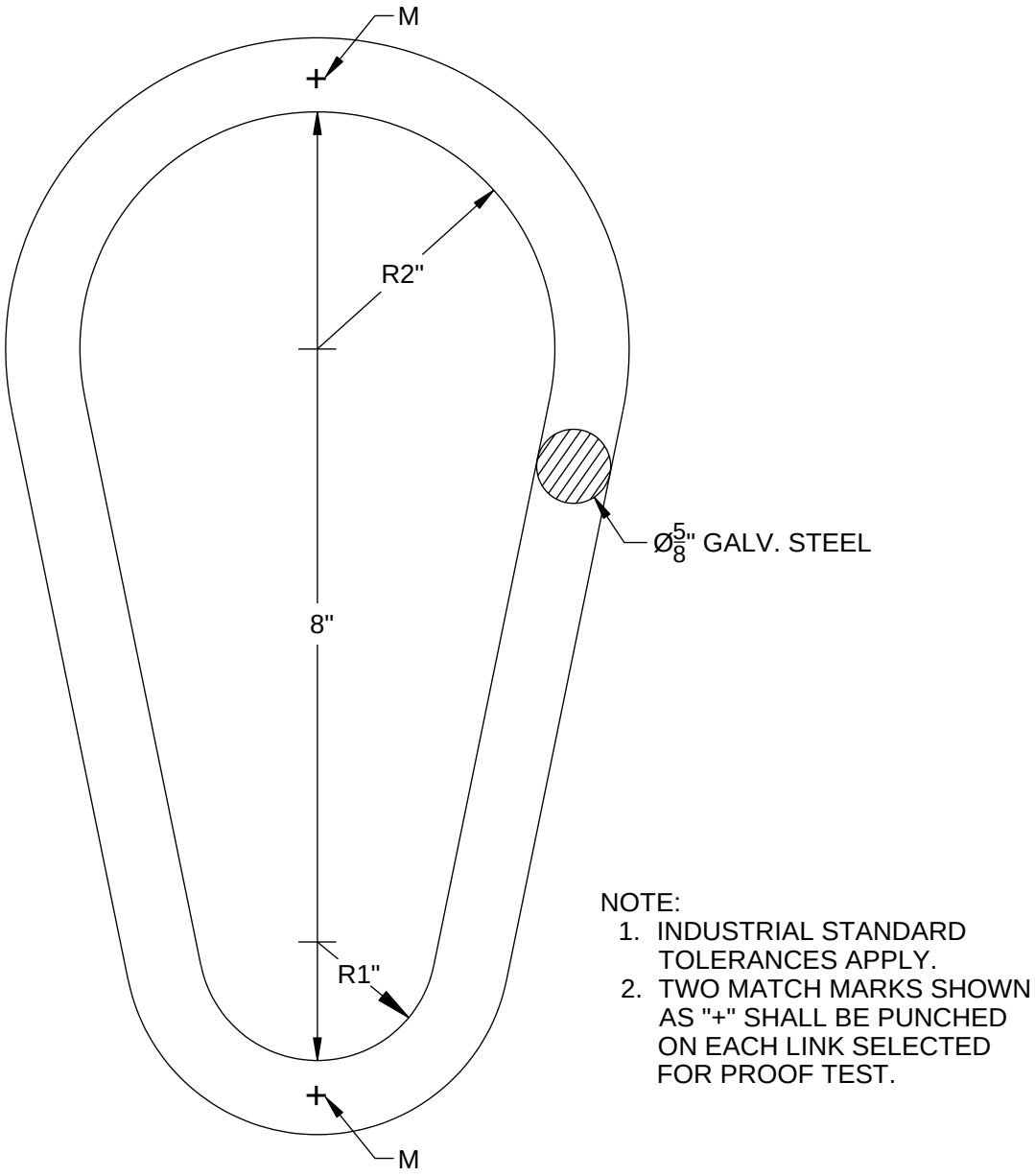


FIGURE 2. Detail of hoist link for nylon webbing sling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

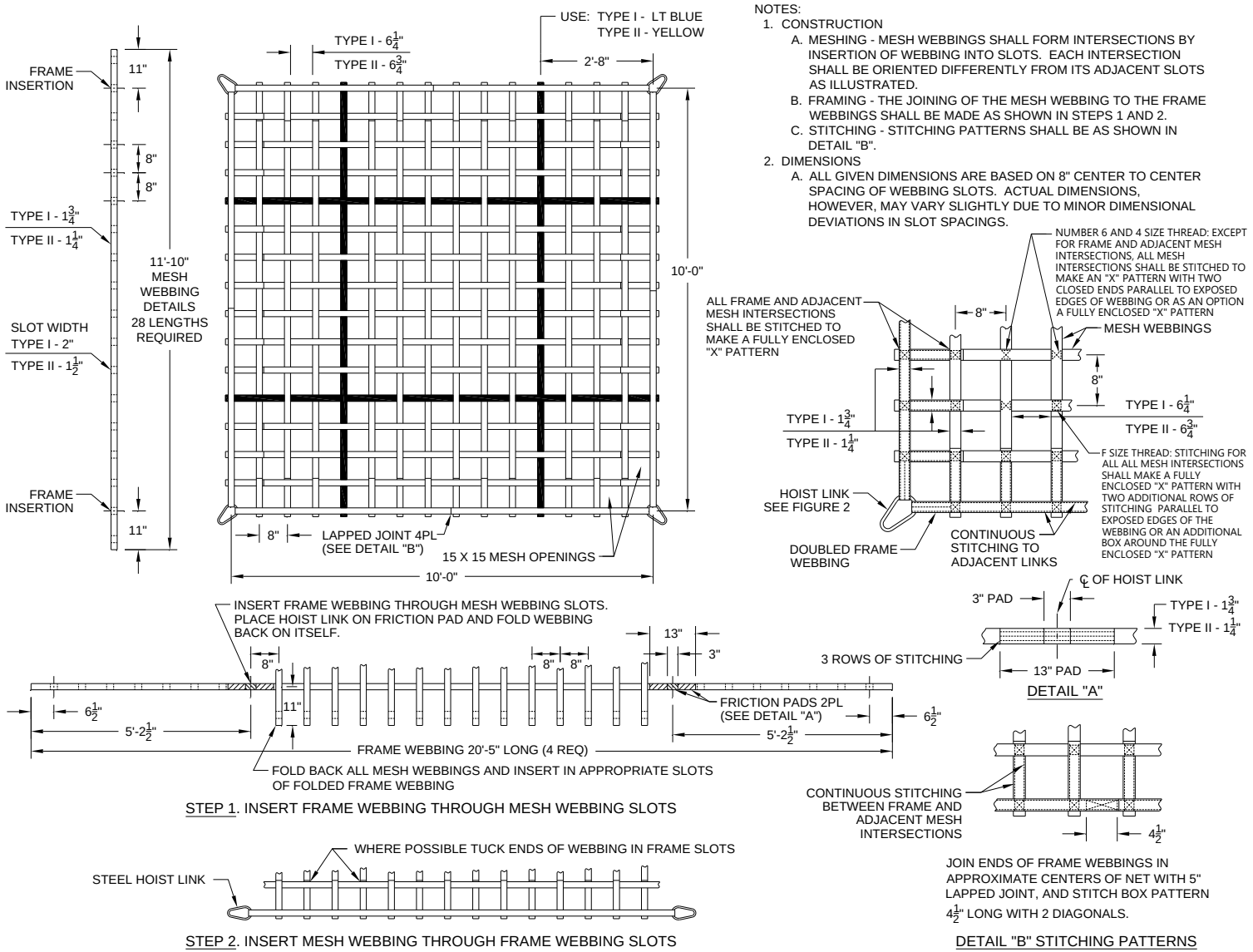
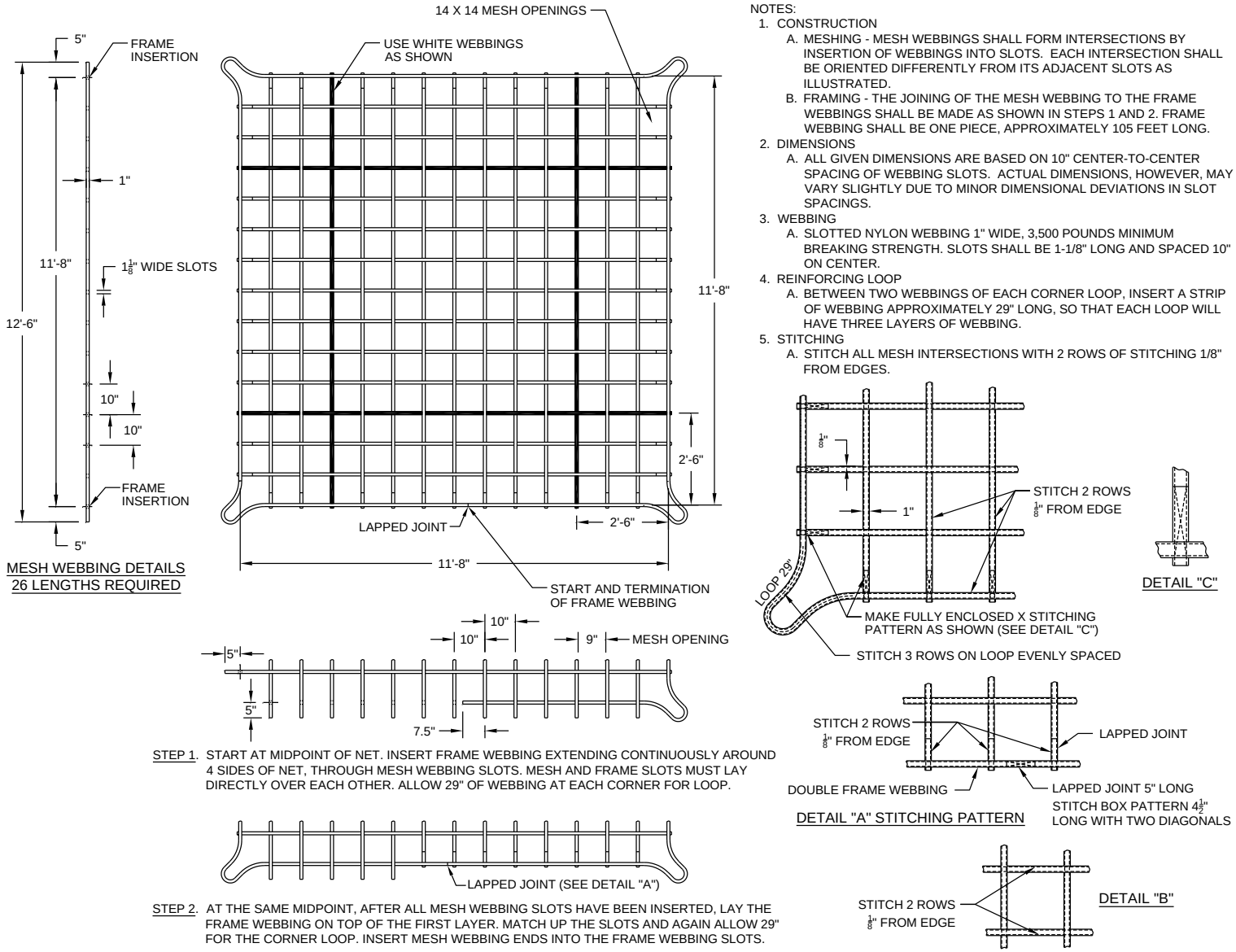


FIGURE 3. Size 1, nylon webbing sling, 10 ft x 10 ft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.



STEP 1. START AT MIDPOINT OF NET. INSERT FRAME WEBBING EXTENDING CONTINUOUSLY AROUND 4 SIDES OF NET, THROUGH MESH WEBBING SLOTS. MESH AND FRAME SLOTS MUST LAY DIRECTLY OVER EACH OTHER. ALLOW 29" OF WEBBING AT EACH CORNER FOR LOOP.

STEP 2. AT THE SAME MIDPOINT, AFTER ALL MESH WEBBING SLOTS HAVE BEEN INSERTED, LAY THE FRAME WEBBING ON TOP OF THE FIRST LAYER. MATCH UP THE SLOTS AND AGAIN ALLOW 29" FOR THE CORNER LOOP. INSERT MESH WEBBING ENDS INTO THE FRAME WEBBING SLOTS.

FIGURE 4. Size 2, nylon webbing cargo sling, 11 ft 8 inches x 11 ft 8 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

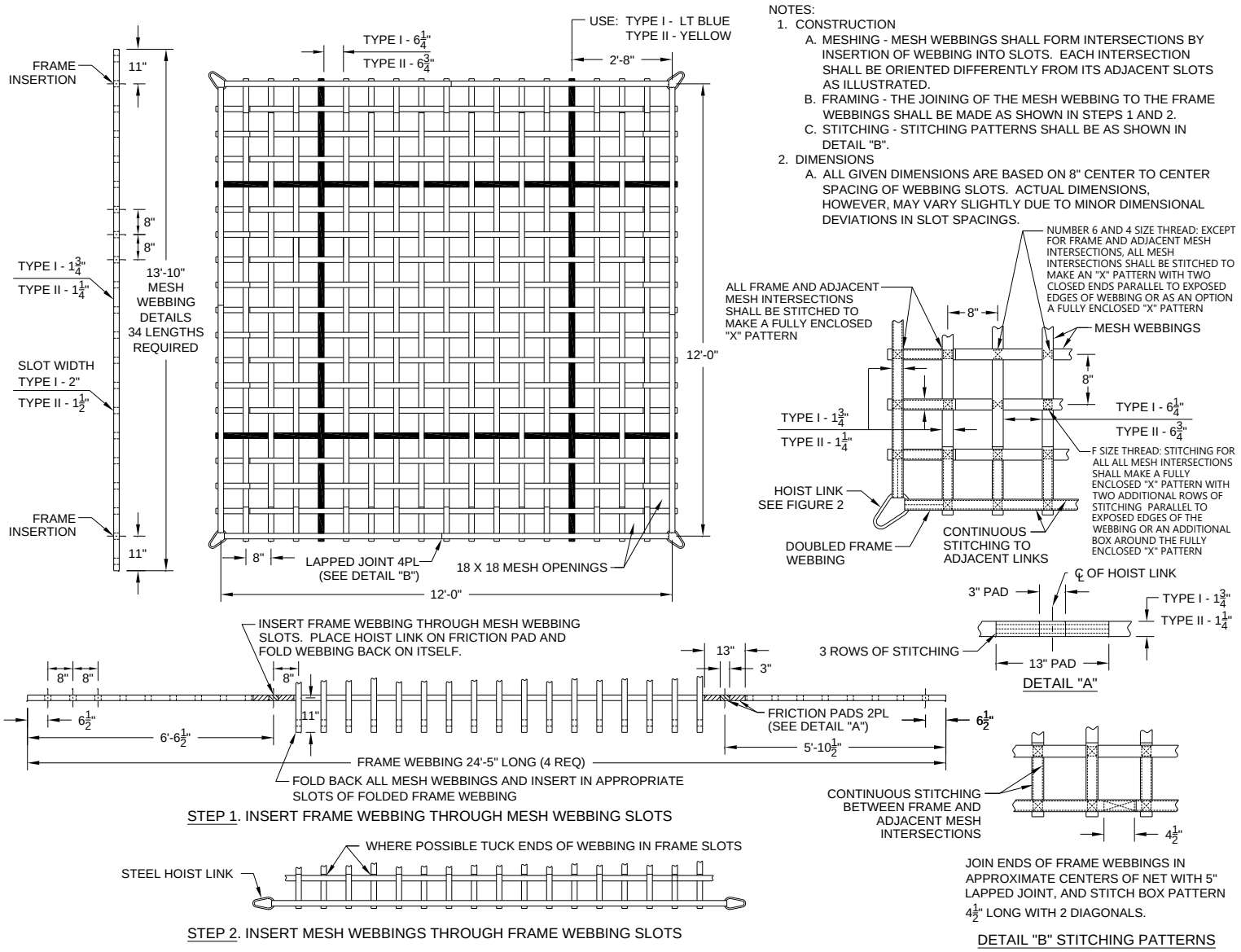


FIGURE 5. Size 3, nylon webbing sling, 12 ft x 12 ft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

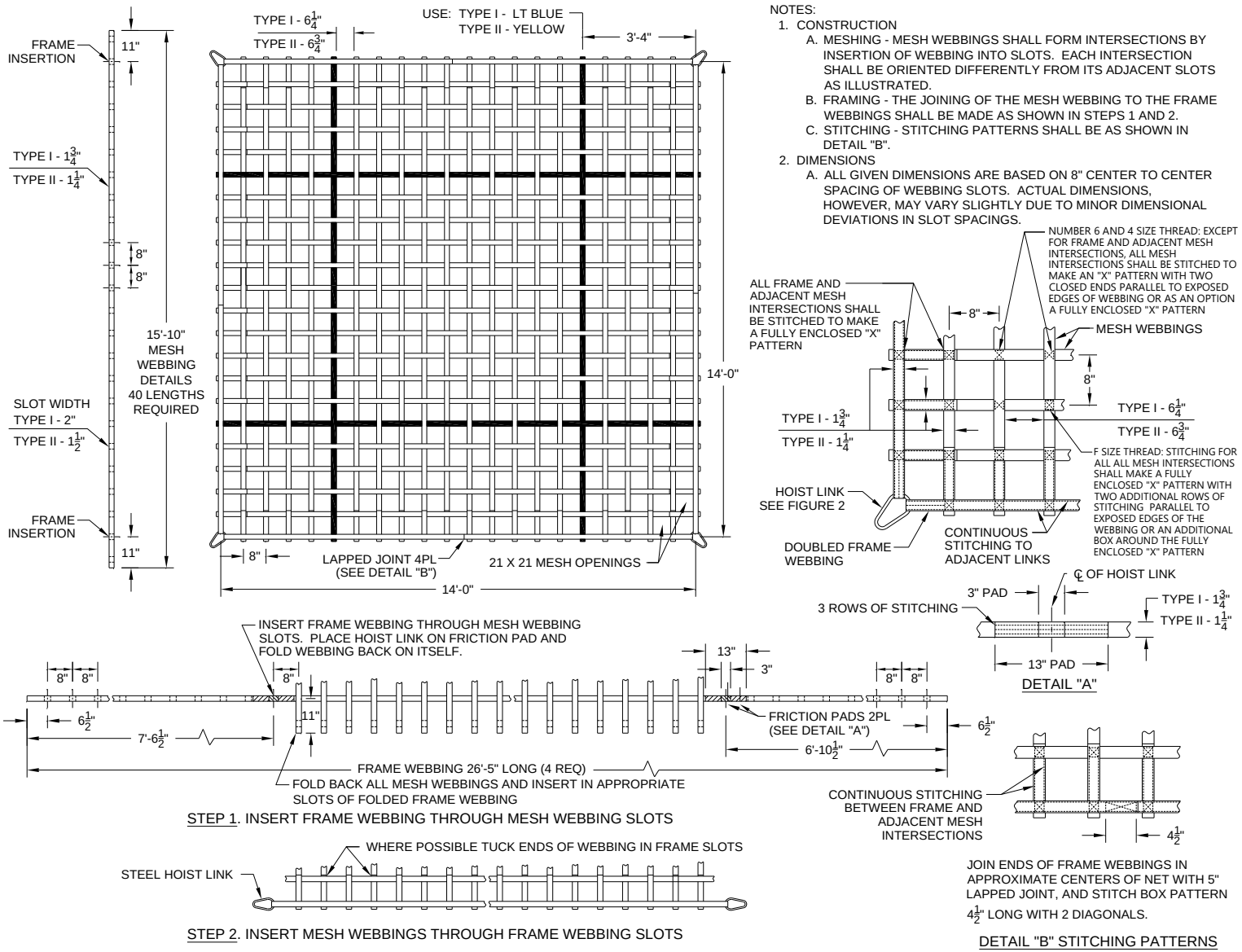


FIGURE 6. Size 4, nylon webbing sling, 14 ft x 14 ft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

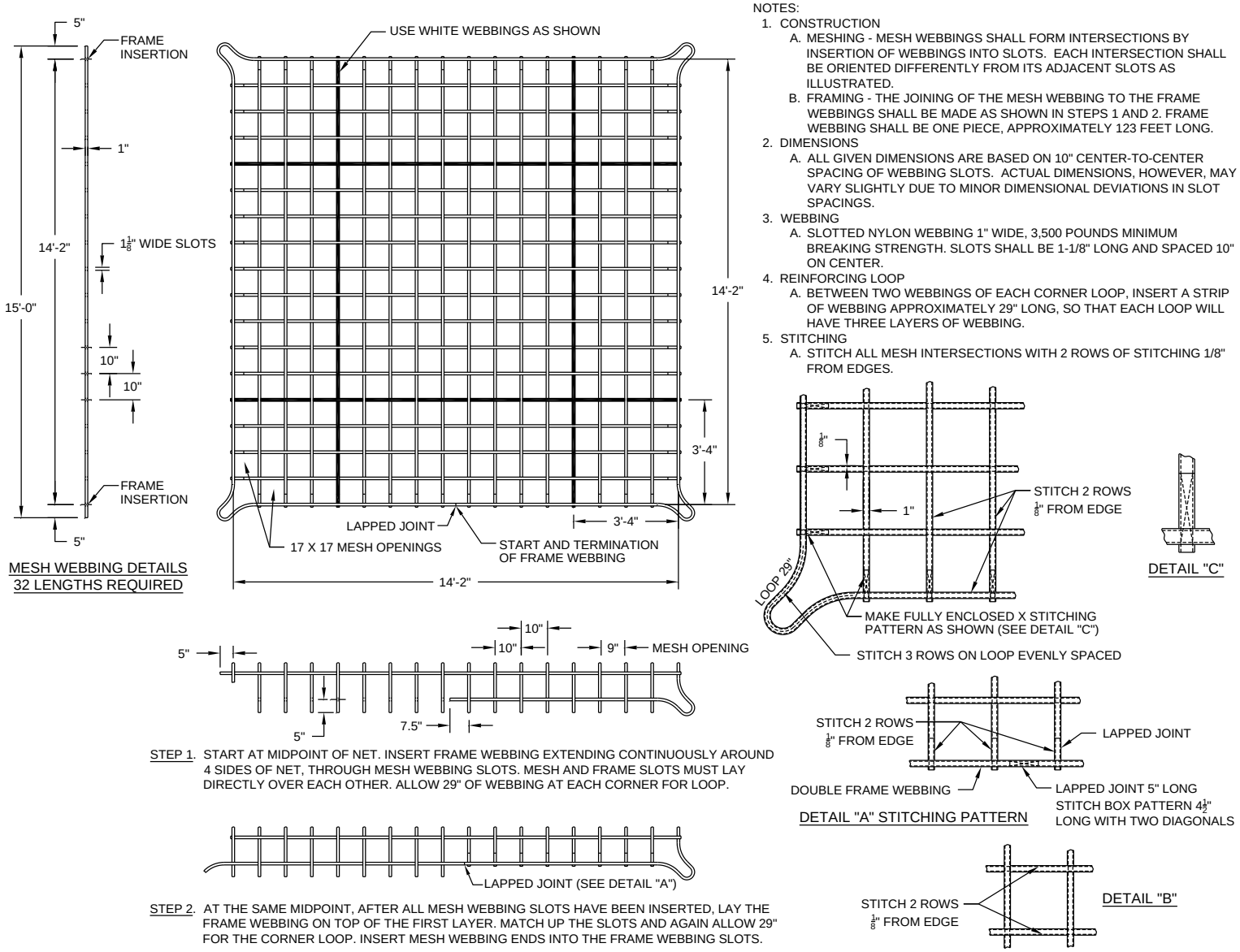


FIGURE 7. Size 5, nylon webbing cargo sling, 14 ft 2 inches x 14 ft 2 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH
Air Force – 184

Review activities:
Army –MT, SM
Navy – OS, YD
DLA – CQ

Preparing activity:
Navy − SH
(Project 3940-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:16Z
Check the source to verify that this is the current version before use.